MILITARY SPECIFICATION

SHIPPING AND STORAGE CONTAINER, ATOMIC WARHEAD SECTION: M480E1 AND M480

1. SCOPE

1.1 This specification covers requirements, quality assurance provisions, and the preparation for delivery criteria for one type of container, entitled Shipping and Storage Container, Atomic Warhead Section: M480E1 and M480 which are used with the Honest John warhead section.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-T-60 - Tape, Pressure-sensitive Adhesive, Waterproof, For Packaging.

MILITARY

MIL-M-48511 - Mount, Resilient.
MIL-M-5021 - Tests; Aircraft Welding Operators Certification.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-810 - Environmental Test Methods.
MIL-STD-1168 - Lot Numbering of Ammunition.
MIL-STD-1169 - Packaging, Packing and Marking for Shipment of Inert Ammunition Components.

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DRAWINGS

ORDNANCE CORPS

8800835 - Front Door Assembly.
8821470 - Block, Pillow.
8821481 - Ring, Support Assembly.
8824426 - Frame, Suspension Assembly.
8826382 - Stud Assembly.
8826385 - Base.
8826386 - Shipping and Storage Container, Atomic Warhead Section: M480E1.
8826387 - Cover, Weldment.
8826388 - Base Assembly.
8826390 - Case Assembly.
8826391 - Side Door Assembly.
8826392 - Mount.
8826401 - Box Assembly, Hardware.
8826402 - Log-Box Assembly.
8826427 - Frame Suspension.
8826428 - Base Frame Assembly.
8831068 - Strap Assembly.
8833479 - Box Assembly, Log.
8833480 - Stud Assembly.
8839515 - Shipping and Storage Container, Atomic Warhead Section: M480.
8839517 - Front Door Assembly.
8839521 - Base.
8839522 - Cover, Weldment.
8839523 - Cover Assembly.
8839524 - Case Assembly.
8839525 - Side Door Assembly.
8839594 - Base Frame and Suspension Assembly.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

3.2 Assembly. The container shall comply with all requirements specified on the drawings (dwgs) 8826386 or 8839515, with all requirements specified herein, and with the requirements of all applicable specifications to the extent specified on the drawing and herein.

3.3 First article testing. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 4.2).

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3.4 Performance characteristics. The container shall be capable of complying with all the requirements specified when loaded with a dummy load. The load shall be properly installed and have the equivalent physical characteristics and weight distribution of the actual load.

3.5 Warhead section compatibility. The container shall be capable of providing mounting for and securely supporting the applicable dummy load specified by the procuring agency. The case shall comply with the following when tested as specified in 4.4.1.

(a) The three access doors, the logbook box and hardware box covers shall be capable of being removed and replaced without undue effort.

(b) The suspension frame support ring shall be capable of being rotated rearward 30 degrees minimum from its normal position.

(c) The case cover shall be capable of being removed from the base by the lifting bars provided and the cover shall balance within 2 degrees of a horizontal plane.

(d) The case shall be capable of being lifted by the bars provided and shall balance within 4 degrees of the horizontal.

3.6 Fork lifting. The loaded container shall be capable of being lifted by a suitable fork type lift truck without evidence of puncture, or broken or bent frame structure as specified in 4.4.2.

3.7 Log-box waterproofness. The log-box shall show no evidence of water leakage when tested as specified in 4.4.3.

3.8 Stacking. The container shall be capable of supporting a load equal to twice the weight of a container loaded with the contents for which it is designed. This load shall be applied in a manner simulating the stacking of two similarly loaded containers. The container shall show no evidence of permanent set or breakage in any of its structural members after this test. The test shall be conducted as specified in 4.4.4.

3.9 Hoisting.

3.9.1 Hoisting container. The container with or without a cargo as specified in 3.5 and loaded to a total weight of 11,200 pounds plus or minus (+) 150 pounds by the addition of weights shall be hoisted or suspended by the lift bars provided. The container shall show no evidence of permanent set or breakage which would compromise its integrity when tested as specified in 4.4.5.1.

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3.9.2 Hoisting base. The base (dwg 8826385 or 8839521) with or without a cargo as specified in 3.5 and loaded to a total weight of 10,650 pounds + 150 pounds by the addition of weights shall be hoisted or suspended by the lift rings provided. The base shall show no evidence of permanent set or breakage which would compromise the integrity of the case when tested as specified in 4.4.5.2.

3.9.3 Hoisting suspension system. The suspension system consisting of the following assemblies, shall be detached from the base and loaded with a dummy load as specified by the procuring agency. This loaded suspension system shall by capable of being hoisted or suspended by the lift rings provided without evidence of permanent set or breakage which would compromise the integrity of the suspension system when tested as specified in 4.4.5.3.

Suspension frame assembly	8826426
Base frame assembly	8826428
Four mounts	8826392
Strap assembly and associated hardware	8831068

3.10 Ground resistance. The resistance between any one of the 12 engaged locking studs on the front door assembly (dwg 8800835 or 8839517) and the unpainted end of any one of the four bushings on the support ring (dwg 8821481) shall not exceed 0.25 ohms when tested as specified in 4.4.6.

3.11 Environmental testing. The container shall comply with MIL-STD-810, visual inspection and failure criteria, section 3. The container shall also comply with 3.5 before and after each environmental test. Testing shall be as specified in 4.4.7. Tests shall be conducted in the order specified.

3.11.1 Handling shock and vibration. The container shall comply with the following requirements when loaded as specified in 3.5. Since the primary function of the container is to protect its contents during storage, transportation and handling; minor dents or creases in non-structural members and cracking of the wooden skids shall not be considered objectional. The resilient mount (dwg 8826392) shall show no evidence of cracks in the elastomer or separation of the elastomer from the steel mounting plates after completion of the tests. The deflection or set of the resilient mount shall not exceed the static load deflection by more than 0.5 inches measured not less than 1 hour after completing the tests. Test shall be as specified in 4.4.7.1. After completion of 3.11.1 thru 3.11.1.5 the container shall comply with the requirements of 3.5.

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3.11.1.1 Edgewise drop. The container shall withstand edgewise dropping and shall show no evidence of loose parts, or broken or permanently set structural members. The contents shall show no evidence of outward damage when tested as specified in 4.4.7.1.1.

3.11.1.2 Cornerwise drop. The container shall withstand cornerwise dropping and shall show no evidence of loose parts, or broken or permanently set structural members. The contents shall show no evidence of outward damage, when tested as specified in 4.4.7.1.2.

3.11.1.3 End impact. The container shall withstand end impacts and shall show no evidence of loose parts, or broken or permanently set structural members. The contents shall show no evidence of outward damage when tested as specified in 4.4.7.1.3.

3.11.1.4 Roll-over impact. The container shall withstand impacts and shall show no evidence of loose parts, or broken or permanently set structural members. The contents shall show no evidence of outward damage when tested as specified in 4.4.7.1.4.

3.11.1.5 Vibration. The container when loaded as specified in 3.5 shall withstand vibration as specified in Table I without evidence of loose parts or fasteners and breakage or cracks in the case structure. The vibration shall be applied in the vertical axis and as specified in 4.4.7.1.5.

Table I

Frequency Range (Hz)	Sweep Time Minutes	Double Amplitude (Inches + 10 Percent Tolerance)	No. of Sweeps
2-10-2	10 ± 0.5	0.4	1
11-15-11	10 ± 0.5	0.25	1
16-20-16	10 ± 0.5	0.13	1
21-60-21	10 ± 0.5	0.06	1

3.12 Records. Records shall be furnished showing that the resilient mounts have been inspected, tested, and accepted in accordance with MIL-M- (PA).

3.13 Welding. Welds shall be reasonably smooth and free of craters. They shall exhibit characteristics of fusion penetration and soundness of weld deposit representative of good welding practice. They shall be free of irregularities which indicate lack of good practice on the part of the operator. All welding fluxes, scale, weld spatter, guide, and basic solutions shall be removed prior to application of any finish coat.

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3.13.1 Welds. Occasional or isolated failure of welds holding the sheets to the structural members shall not be considered objectionable providing it meets the requirements of a through e.

a. (1) No more than three consecutive plug welds shall fail.
   (2) No more than 25 percent of the total welds in line on a straight framing member shall fail.
   (3) No more than a total of 50 plug weld failures occur on any one cover assembly (8826388 or 8839515, as applicable).
   (4) No more than a total of 20 plug weld failures occur on any one base assembly (8826388 or 8839515, as applicable).

b. Plug and spot welds which permit the sheet to separate from the framing member at the weld shall be considered unsatisfactory.

c. Omitted welds shall be considered weld failures.

d. The number of welds omitted shall be determined by measuring the distance between centerline of adjacent acceptable welds and multiplying by 0.62 and subtracting 1 from the product. The resultant answer will be rounded off to the nearest whole number.

e. In such case where the sheet metal is spotwelded, all failures of spotwelds are to be considered as plug weld failures.

3.13.2 Welding operators. All fusion welding shall be performed by operators who are currently certified as Class B welders in accordance with MIL-T-5021.

3.14 Paint adhesion. No bare spots shall be visible when tested in accordance with 4.4.8.

3.15 Workmanship.

3.15.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.15.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.15.3 Burrs. Parts shall be free from burrs.

3.15.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.15.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection.

4.2.1 Quantity to be submitted. When the contract requires first article inspection, the contractor shall submit 1 complete assembly (see dwgs 8826386 or 8839515) for first article testing to the testing facility designated in the invitation for bids or request for proposal.

4.2.2 Inspection and testing to be performed. The first article inspection sample shall be subjected to the inspections and tests specified in Table II. Tests shall be conducted in order specified in Table II.

Table II

First article inspection and tests

Inspection or Test	Dest	Non-Dest
1. Classification of characteristics (see 4.3.2.1)		X
2. Warhead section compatibility (see 3.5)		X
3. Ground resistance (see 3.10)		X
4. Fork lifting (see 3.6)		X
5. Stocking (see 3.8)		X
6. Hoisting (see 3.9)		X
7. Log-box waterproofness (see 3.7)		X
8. Vibration (see 3.11.1 and 3.11.1.5)	X	
9. Handling shock (see 3.11.1 thru 3.11.1.4)	X	
10. Paint adhesion (see 3.14)		X

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4.2.3 Inspection equipment. The inspection and testing specified in 4.2.2 shall be accomplished by using the gages and test equipment specified in 4.3.4.1.

4.2.4 Rejection. If any item fails to comply with any of the applicable requirements, the first article sample shall be rejected.

4.3 Quality conformance inspection.

4.3.1 Inspection lot formation. All items in each inspection lot shall be from one production line, one shift, one batch of products, one manufacturer or raw material or components, one production method or manufacturing operation. The entire lot shall be submitted as a unit for inspection.

4.3.2 Product inspection examination. Critical characteristics shall require 100 percent inspection of all items in the inspection lot. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristic table (see Table III and Table IV), and inspected for all characteristics in the applicable classification. Any characteristics in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed as long as all characteristic inspections are performed.

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Table III
Major Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-16	all	28-29	23	52-56	31	132-156	39
17	16	30-31	24	57-62	32	157-191	40
18	17	32-34	25	63-69	33	192-244	41
19-20	18	35-36	26	70-76	34	245-334	42
21	19	37-40	27	77-86	35	335-519	43
22-23	20	41-43	28	87-97	36	520-1120	44
24-25	21	44-47	29	98-112	37	over 1120	45
26-27	22	48-51	30	113-131	38		

Table IV
Minor Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-9	all	14-16	12	28-33	16	77-121	20
10	9	17-18	13	34-41	17	122-273	21
11	10	19-22	14	42-54	18	over 273	22
12-13	11	23-27	15	55-76	19		

4.3.2.1 Classification of characteristics (see 3.2). Critical, major and minor characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

4.3.2.1.1 Box assembly, log (see dwg 8833479 covering a detail of dwg 8839515, or see dwg 8826402 covering a detail of dwg 8826386, as applicable).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Evidence of poor welding (see 3.13)	Visual
	102. Finish inadequate or incorrect	Visual
	103. Excessive gap between gasket and sides and ends of cover	Visual
	104. Gasket missing, loose or incorrect	Visual/Manual
	105. Any catch fails to function properly	Visual/Manual
	106. Cover fits loosely	Visual/Manual
Minor:		
	201. Evidence of poor workmanship (see 3.15)	Visual

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4.3.2.1.2 Base frame assembly (see dwg 8826428 covering a detail of dwg 8839515 or dwg 8826386, as applicable).

Categories	Characteristics	Method of Inspection
Critical:		
	1. Evidence of poor welding (see 3.13)	Visual
Major:		
	101. Location and diameter of four holes in side of aft bracket	Gage
	102. Location and diameter of four holes in bottom of aft bracket	Gage
	103. Location and diameter of four holes in side of forward bracket	Gage
	104. Location and diameter of four holes in bottom of forward bracket	Gage
	105. Location of threaded holes	Gage
	106. Pitch diameter of threads (2 places)	Gage
	107. Angle missing or incorrect	Visual
	108. Paint in threaded holes	Visual
	109. Handle missing, loose or incorrect	Visual/Manual
Minor:		
	201. Evidence of poor workmanship (see 3.15)	Visual
	202. Finish inadequate or incorrect	Visual

4.3.2.1.3 Frame suspension (see dwg 8826427 covering a detail of dwg 8839515 or 8826386, as applicable).

Categories	Characteristics	Method of Inspection
Critical:		
	1. Evidence of poor welding (see 3.13)	Visual
Major:		
	101. Diameter and location of three holes in mounting plates	Gage
	102. Width at rear end	Gage
	103. Width at forward end	Gage
Minor:		
	201. Evidence of poor workmanship (see 3.15)	Visual
	202. Finish inadequate or incorrect	Visual

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4.3.2.1.4 Block, pillow (see dwg 8821470 covering a detail of dwg 8839515 or dwg 8826386, as applicable).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Diameter of two small holes	Gage
	102. Pitch diameter of thread	Gage
	103. Minor diameter of thread, min	Gage
	104. Diameter of large hole	Gage
	105. Location of large hole with base	Gage
Minor:		
	201. Evidence of poor workmanship (see 3.15)	Visual
	202. Finish inadequate	Visual

4.3.2.1.5 Ring, support assembly (see dwg 8821481 covering a detail of dwg 8839515 or dwg 8826386, as applicable).

Categories	Characteristics	Method of Inspection
Critical:		
	1. Evidence of poor welding (see 3.13)	Visual
Major:		
	101. Location and diameter of four holes in forward surface	Gage
	102. Bracket missing or incorrect	Visual
	103. Gusset or stiffener missing	Visual
	104. Location of mounting pins	Gage
	105. Diameter of mounting pins	Gage
Minor:		
	201. Evidence of poor workmanship (see 3.15)	Visual
	202. Paint in threaded holes or on ends of bushing	Visual

4.3.2.1.6 Frame, suspension, assembly (see dwg 8826426 covering a detail of dwg 8839515 or dwg 8826386, as applicable).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	

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Major:
101. Rubber pad loose, missing or incorrect.....Visual/Manual
102. Pillow blocks loose or assembled incorrectly to suspension frame.....Visual/Manual
103. Hardware loose or missing.....Visual/Manual
104. Ring binds (prior to adjustment of set screws).....Manual
105. Set screws fail to hold ring securely.....Manual

Minor:
201. Evidence of poor workmanship (see 3.15).....Visual
202. Protective coating on pins missing, inadequate, or incorrect.....Visual

4.3.2.1.7 Strap assembly (see dwg 8831068 covering a detail of dwg 8839515 or dwg 8826386, as applicable).

Categories	Characteristics	Method of Inspection

Critical:
1. Evidence of poor welding (see 3.13).....Visual

Major:
101. Length between holes at maximum extension.....Gage
102. Length between holes at minimum extension.....Gage
103. Any part missing or damaged.....Visual
104. Thread damaged.....Visual
105. Felt pad missing, loose, or damaged.....Visual/Manual
106. Screw fails to mate properly with trunnion or strap assembly adjuster....Visual/Manual

Minor:
201. Evidence of poor workmanship (see 3.15).....Visual

4.3.2.1.8 Base frame and suspension assembly (see dwg 8839594 covering a detail of dwg 8839515 or see dwg 8826385 covering a detail of dwg 8826386).

Categories	Characteristics	Method of Inspection

Critical: None defined.

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Major:
101. Any assembly loosely or incorrectly assembled.....Visual/Manual
102. Hardware loose or missing.....Visual/Manual

Minor:
201. Evidence of poor workmanship (see 3.12).....Visual

4.3.2.1.9 Box assembly, hardware (see dwg 8826401 covering a detail of dwg 8839515 or dwg 8826386, as applicable).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Location and diameter of hole in body (2 places)	Gage
102.	Strike missing or incorrect	Visual
103.	Evidence of poor welding (see 3.13)	Visual
104.	Catch missing or incorrect	Visual
105.	Finish inadequate or incorrect	Visual
106.	Catch fails to function properly or hold lid tightly	Visual/Manual
Minor:		
201.	Evidence of poor workmanship (see 3.15)	Visual

4.3.2.1.10 Front door assembly (see dwg 8839517 covering a detail of dwg 8839515 or see dwg 8800835 covering a detail of dwg 8826386, as applicable).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Backing plate missing or incorrect	Visual
102.	Evidence of poor welding	Visual
103.	Stud missing, incorrect or incorrectly assembled	Visual/Manual
104.	Handle loose, missing or incorrect	Visual/Manual
Minor:		
201.	Evidence of poor workmanship (see 3.15)	Visual

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4.3.2.1.11 Side door assembly (see dwg 8839525 covering a detail of dwg 8839515 or dwg 8826391 covering a detail of dwg 8826386, as applicable).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Chain with "S" hook missing or incorrectly assembled	Visual
102.	Evidence of poor welding (see 3.13)	Visual
103.	Stud missing, incorrect or incorrectly assembled	Visual/Manual
Minor:		
201.	Evidence of poor workmanship (see 3.15)	Visual

4.3.2.1.12 Cover, weldment (see dwg 8839522 covering a detail of dwg 8839515, or see dwg 8826387 covering a detail of dwg 8826386).

Categories	Characteristics	Method of Inspection
Critical:		
1.	Evidence of poor welding (see 3.13)	Visual
Major:		
101.	Major diameter of thread on stud, max	Gage
102.	Thread on stud damaged	Visual
103.	Angle missing or incorrect (8)	Visual
104.	Marking missing, mislocated, or incorrect	Visual
105.	Receptacle missing or incorrect (12)	Visual
Minor:		
201.	Evidence of poor workmanship (see 3.15)	Visual

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4.3.2.1.13 Cover assembly (see dwg 8839523 covering a detail of dwg 8839515 or see dwg 8826388 covering a detail of dwg 8826386).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Stacking bar missing or incorrect	Visual
	102. Chain missing or assembled incorrectly	Visual
	103. Quickie washer connot be moved to unlock by hand	Manual
Minor:		
	201. Grease on stud thread missing	Visual

4.3.2.1.14 Base assembly (see dwg 8839521 covering a detail of dwg 8839515 or see dwg 8826385 covering a detail of dwg 8826386 as applicable).

Categories	Characteristics	Method of Inspection
Critical:		
	1. Evidence of poor welding (see 3.13)	Visual
Major:		
	101. Bumper or skid cracked or split	Visual
	102. Attaching nuts or screws extend beyond bumper or skid outer surface	Visual
	103. Protective finish inadequate	Visual
	104. Handle missing or incorrect	Visual
	105. Hardware loose or missing	Visual/Manual
Minor:		
	201. Evidence of poor workmanship (see 3.15)	Visual

4.3.2.1.15 Stud assembly (see dwg 8833480 covering a detail of dwg 8839515 or see dwg 8826382 covering a detail of dwg 8826386 as applicable).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Major diameter of thread, max	Gage
	102. Pitch diameter of thread	Gage
Minor:	None defined.	

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4.3.2.1.16 Case assembly (see dwg 8839524 covering a detail of dwg 8839515 or see dwg 8826390 covering a detail of dwg 8826386 as applicable).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Any component part missing or incorrectly assembled	Visual
102.	Threads not lubricated	Visual
Minor:		
201.	Evidence of poor workmanship (see 3.15)	Visual

4.3.2.1.17 Shipping and storage container, atomic warhead section: M480E1 or M480 (see dwg 8826386 or 8839515 as applicable).

Categories	Characteristics	Method of Inspection
Critical:		
1.	Punctures or excessive dents in panel	Visual
Major:		
101.	Jumper missing or incorrect	Visual
102.	Any component part missing, damaged or incorrectly assembled	Visual
103.	Bag missing or incorrect	Visual
104.	Lubricant missing from swivel lock or bearings of caster	Visual
105.	Caster swivel lock fails to function	Visual/Manual
106.	Hardware loose or missing	Visual/Manual
107.	Log box difficult to remove	Manual
Minor:		
201.	Marking missing or incorrect	Visual
202.	Evidence of poor workmanship (see 3.15)	Visual

4.3.3 Testing. All tests shall be performed. Tests shall be performed in the order specified below:

	Test	Test Classification
a.	Log-box waterproofness	Major
b.	Ground resistance	Major
c.	Handling shock and vibration	Major
d.	Paint adhesion	Minor

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Of these tests, c is considered destructive and samples so tested shall not be returned to the lot. Destructive samples are over and above the deliverable quantity of items required by the contract.

4.3.3.1 Log-box waterproofness, ground resistance, and paint adhesion. Each container shall be tested in accordance with 4.4.3, 4.4.6 and 4.4.8. If any item fails to comply with 3.7, 3.10 and 3.14, the item shall be rejected and removed from the lot.

4.3.3.2 Handling shock and vibration. One sample container of each 300 produced shall be subjected to this test (see 4.4.7). If the sample fails to comply with the applicable requirement, all units produced subsequent to the discovery of the failure shall not be shipped until the cause of failure has been determined and the units withheld from shipment have been corrected.

4.3.4 Acceptance inspection equipment.

4.3.4.1 Major characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions and calibration instructions designed or specified by the contractor. The contractor shall obtain approval of such designs, operating instructions and calibration instructions prior to use on the contract from Commander, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SARPA-QA-N-I as specified on DI-R-1714 (see 6.1.2).

4.3.4.2 Minor characteristics and tests. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used.

4.4.1 Warhead section compatibility. The container shall be loaded as specified. The installation of the load shall be accomplished without damage or the use of excessive force. Requirements shall be as specified in 3.5.

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4.4.2 Fork lifting. The container shall be lifted clear of another container (or equivalent) when stacked, lowered to the floor by a fork truck having forks of the following description:

Length of 70 inches minimum for end lifting, and

Length of 32 inches minimum for side lifting,

overall width adjustable within range of 30 to 45 inches. The fork truck shall be cleared of the test case, resume lifting position, and restack. This procedure may be started with the case on the floor. This procedure shall be followed in lifting once from each side and the aft end as indicated by the marking for the center of balance, and from the aft end.

4.4.3 Log-box waterproofness. The log-box assembly 8826402 or 8833479 when removed from the base assembly, shall be submerged in a water bath to a depth of not less than 1 inch for 2 minutes minimum (min). Immediately upon removal from the water bath inspect interior for water entry.

4.4.4 Stacking test. A load of 5400 pounds ± 50 pounds or two loaded cases shall be placed on top of the case in such a manner as to simulate stacking of two additional loaded cases on the first.

4.4.5 Hoisting.

4.4.5.1 Hoisting case. Additional weight shall be added to the base to bring the total weight of the case to 11,200 pounds ± 150 pounds and so distributed that the center of balance shall be a minimum of 70.0 inches and a maximum of 82.0 inches from the forward end of the case, exclusive of wood bumpers. A pair of slings at an angle of 30 degrees ± 5 degrees to the top of the case and extending from a diagonally opposite pair of lift bars to a single suspension point shall be used to hoist or suspend the loaded case. The test shall be repeated using the remaining pair of diagonally opposite lift bars. Care should be exercised to restrict hoisting accelerations to the smallest practicable minimum.

4.4.5.2 Hoisting base. With the cover removed, additional weight shall be applied to the base to bring the total weight to 10,650 pounds ± 150 pounds and so distributed that the center of balance shall be minimum of 74.0 inches and a maximum of 86.0 inches from the forward end of the base, exclusive of wood bumpers. A pair of slings extending from a diagonally opposite pair of lift rings to a single suspension point and at an angle of 45 degrees ± 5 degrees from the plane of the base or at the minimum angle necessary to avoid interference with the load section shall be used to hoist or suspend the loaded base. The test shall be repeated using the remaining pair of diagonally opposite lift rings. Care should be exercised to restrict hoisting accelerations to the smallest practicable minimum.

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4.4.5.3 Hoisting suspension system. The suspension system loaded and detached from the base shall be hoisted or suspended by a pair of slings extending from diagonally opposite lift rings to a single suspension point and at the minimum angle to the plane of the suspension base frame to avoid interference with the load section or suspension structure. The test shall be repeated using the remaining pair of diagonally opposite lift rings.

4.4.6 Ground resistance. The resistance measurement shall be made with an instrument whose current output is limited to 20 milliamperes maximum. Requirements shall be as specified in 3.10. Note: Installation of cargo may be omitted for this test.

4.4.7 Environmental testing (see 3.11).

4.4.7.1 Handling shock and vibration. Prior to testing the distance between a point on the circular mounting plate of the resilient mount (dwg 8826392) or a point on the suspension frame assembly (dwg 8826426) close to the mount and a fixed point below the base (dwg 8826385 or 8839521) or the base frame assembly (dwg 8826428) shall be measured. Not less than 1 hour after completion of the tests, the distance between the same two points shall be measured. The difference between the two measurements shall not exceed 0.5 inches; all four mounts shall be so measured. Following these tests the assembly shall be tested as specified in 4.4.1.

4.4.7.1.1 Edgewise drop test. Support the loaded case on one end of its base on a wood sill 5 ± 0.5 inches high and placed normal to the skids. The opposite end of the case shall be raised and allowed to drop freely once from 12, 24 and 36 inch heights in the order listed. The three drops shall be made on concrete or a similar hard surface with a tolerance of ± 0.5 inch applying to all heights. Apply the test to each end of the case.

4.4.7.1.2 Cornerwise drop. Support the loaded case at one corner of the base on a block 5 ± 0.5 inches high. Support the other corner of the same end on a block 12 ± 0.5 inches high. The opposite end of the case shall be raised and allowed to drop freely once from 12, 24 and 36 inch heights in the order listed. The three drops shall be made on concrete or a similar hard surface with a tolerance of ± 0.5 inch applying to all heights. The test shall be done on one diagonally opposite corner of each end.

4.4.7.1.3 Horizontal and impact test. The loaded case shall be packed in a fixture which will allow free axial movement and cause the base end of the case to strike against a solid flat vertical abutment having a wooden face 2 inches minimum thickness. The direction of motion shall be at 90 degrees ± 5 degrees with the impact surface and the velocity at impact shall be from 9 to 12 feet per second. The test shall be made once at each end.

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4.4.7.1.4 Roll-over test. The loaded case, erect on its base on a hard, level floor, shall be tipped slowly sideways until it falls freely and by its own weight on its side. This process shall be repeated with falls from the side to the top to the other side, and from the other side to its base, thus completing one revolution.

4.4.7.1.5 Vibration test. Place the loaded case on a suitable fixture and vibrate it in the vertical axis accordin to the schedule of Table I.

4.4.8 Paint adhesion. This test shall be performed on each container after the paint has been allowed to dry for a minimum of 24 hours. Adhesive tape 3/4" width, conforming to PPP-T-60, Type II of III, grade A or B shall be used for this test. Firmly press a two inch length or longer piece of tape onto the painted surface of the container. Press out air bubbles underneath tape. Allow 10 seconds for the container to return to room temperature. Firmly grip a free end of the tape. Rapidly strip the tape from the surface by pulling the tape back to back upon itself 180 degrees. Closely observe for bare spots around and under affected area. Disregard flecks of paint on tape. However, removal of paint exposing any coating or base metal shall be cause for rejection.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A. The container shall be packaged in accordance with dwgs 8826386 or 8839515, and MIL-STD-1169 using the specified moisture barrier.

5.1.2 Level C. The container shall be packaged in accordance with dwgs 8826386 or 8839515, and MIL-STD-1169 except that the moisture barrier may be omitted.

5.2 Packing.

5.2.1 Level A and C. The container shall be packed in accordance with dwgs 8826386 or 8839515, and MIL-STD-1169.

5.3 Marking. Marking shall be as specified in accordance with dwgs 8826386 or 8839515, and MIL-STD-1169.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

6.1.1 Procurement requirements.

a. Title, number, and date of this specification.

b. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.1.2 Contract data requirements (see 4.3.4.1). Data specified in 4.3.4.1 will be delivered as identified on a numbered DD Form 1664 when required on a DD Form 1423 incorporated in the contract.

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6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment (see 4.3.4.1). The contractor shall obtain approval from the Commanding Officer, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SARPA-QA-N-I.

6.4 Supersession data. This specification supersedes the requirements of Purchase Description PA-PD-1686 Amendment 3 dated 8 June 1965.

Custodian:
Army - PA

Preparing activity:
Army - PA

Project No. NUOR-A019

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